

1.1 Introduction

A meeting recording is hard to use as a record. Speech recognition stumbles on technical terms and names, several people speak at once, and spoken decisions are often tentative or vague. This work presents a system that turns a raw meeting recording into a transcript and a structured record containing a summary, minutes, decisions, and action items.

The main requirement is faithfulness. The system corrects recognition errors, but it never adds an owner, deadline, or decision that the participants did not state. It accepts English and Hindi recordings; Hindi is converted to an English transcript before the shared refinement and documentation stages.

1.2 Literature Review

1.2.1 Rennard et al. [1]

Rennard et al. [1] survey abstractive meeting summarization and list its main difficulties: disfluent speech, multiple speakers, long conversations, and factual consistency. They organize existing work into three steps: interpret the conversation, transform it into a structured representation, and generate the summary. They also stress the value of speaker information, topic segmentation, and context for finding decisions. This is why our system places a Transformation layer between the refined transcript and the final documentation.

1.2.2 Du et al. [2]

Du et al. [2] study speaker diarization in multi-party meetings, focusing on overlapping speech. Their Speaker Overlap-aware Neural Diarization (SOND) model represents overlapping speakers explicitly to improve attribution. We face the same attribution problem and use PyAnnote Community-1 for diarization, aligned with timestamped ASR output.

1.2.3 Jin et al. [3]

Jin et al. [3] propose CFAS, a summarization framework that identifies topics and consensus points and gives more weight to utterances that express agreement or decisions. Following this idea, our Transformation stage includes consensus analysis. It separates confirmed agreements from proposals, contested points, and unresolved items, so a decision-sounding sentence is not automatically treated as final.

1.2.4 Inoue et al. [4]

Inoue et al. [4] present the Meeting Decision Tracker, which finds decision-related utterances and rewrites them as self-contained statements, since spoken decisions often depend on surrounding dialogue. We use the same idea: the Transformation stage uses neighbouring segments to make each decision self-contained while keeping links to the original evidence.

1.3 Methodology

The system is a sequential pipeline of four steps: transcription with speaker diarization, transcript refinement, evidence-based processing, and documentation. Each step reads the output of the previous one, and all intermediate results are kept so any output can be traced back to the recording.

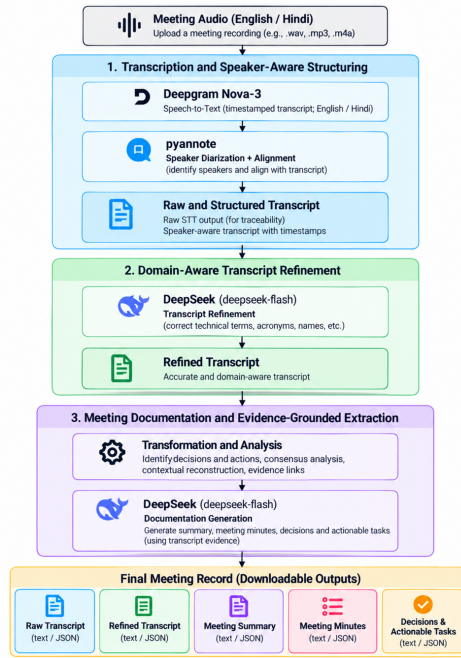


Figure 1: Overview of the PULSE processing pipeline.

1.3.1 Transcription and Speaker-Aware Structuring

Deepgram Nova-3 produces a timestamped transcript. This raw output is stored unchanged as the reference for everything that follows. PyAnnote Community-1 then labels who spoke when, and the result is aligned with the ASR segments. Each aligned segment carries a speaker, start time, end time, and text. Keeping the raw and aligned versions separate makes it possible to see exactly what later stages changed.

1.3.2 Domain-Aware Transcript Refinement

DeepSeek (deepseek-flash) corrects likely recognition errors in technical terms, acronyms, names, and identifiers. It does not summarize or interpret. The refinement is deliberately conservative: speaker labels, timestamps, ordering, numbers, negation, uncertainty, questions, proposals, and commitments must stay as they were. The model returns the refined transcript together with a record of each change.

A deterministic validation layer checks every proposed change against the raw segment before accepting it. Edits that are unsupported, or that alter numbers or negation, are rejected.

1.3.3 Meeting Documentation and Evidence-Grounded Extraction

Before documentation is written, the Transformation layer finds candidate decisions and action items and links each to its transcript segment IDs. It runs consensus analysis to tell confirmed agreements apart from proposals, tentative statements, contested points, and unresolved items. Where a decision only makes sense in context, contextual reconstruction rewrites it as a self-contained statement and keeps the references to the original segments.

This evidence-linked output and the refined transcript go to a second DeepSeek (deepseek-flash) call. Its job is different from refinement: it writes the summary, organized minutes, decisions, and action tasks. Owners and deadlines appear only when the meeting explicitly stated them. Proposals, recommendations, conditional statements, and unresolved discussions are not turned into decisions or assignments.

The final record is saved in readable and machine-readable formats. Every decision and action item keeps references to its supporting segments, so a user can check the speaker, timestamp, and wording behind it.

1.4 Interactive Application

The application runs the whole pipeline from one interface. The user uploads a recording and receives the final meeting record without running any pipeline component by hand.

- **Pipeline status:** each stage (transcription, diarization, alignment, refinement, transformation, documentation) is shown as pending, running, done, or failed.
- **Transcript review:** the raw and refined transcripts can be viewed side by side, with the changes made during refinement highlighted.
- **Meeting documentation:** the summary, minutes, decisions, and action items are shown in separate tabs.
- **Evidence review:** selecting a decision or action item shows its supporting segments with speaker, timestamp, and transcript text.
- **Warnings:** items with ambiguous or weak evidence are flagged for review and not presented as certain.
- **Downloads:** the raw transcript, refined transcript, summary, minutes, decisions, action items, and the full structured record (JSON) can each be downloaded separately.
- **Error handling:** unsupported, empty, unreadable, or failed recordings produce a clear message that names the failed stage and offers a retry. Raw system errors are not shown to the user.
- **Ask PULSE:** the user can ask natural-language questions about a processed meeting. Answers come only from the transcript and documentation and cite the segments they use, so decisions and action items can be found without searching the whole transcript.

1.5 Application Interface

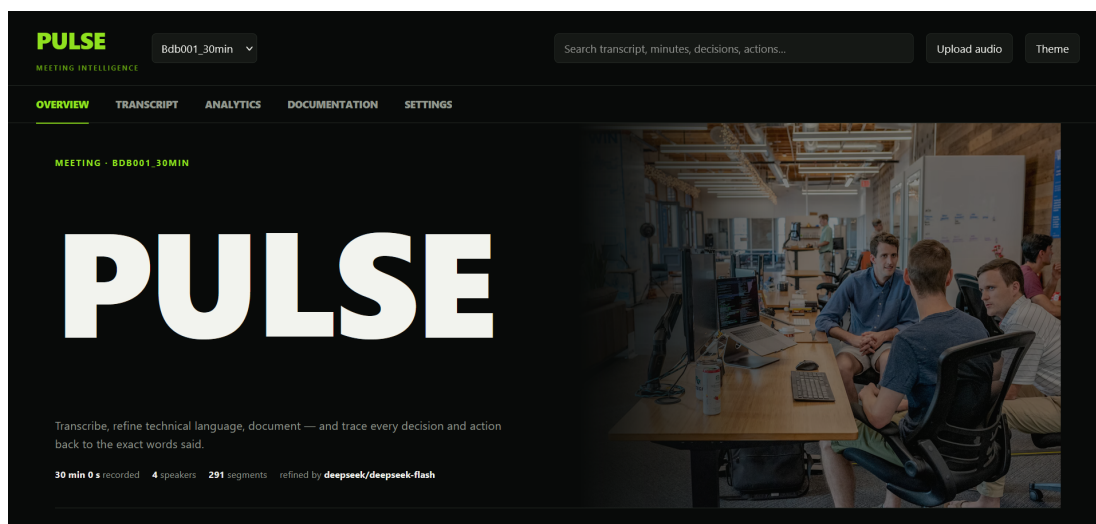


Figure 2: Overview screen.

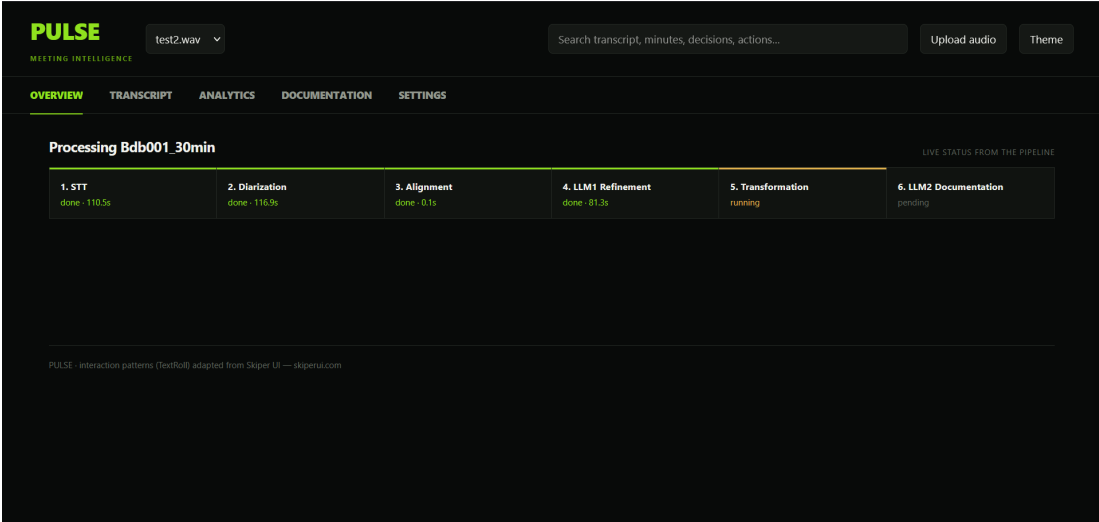


Figure 3: Live pipeline status during processing.

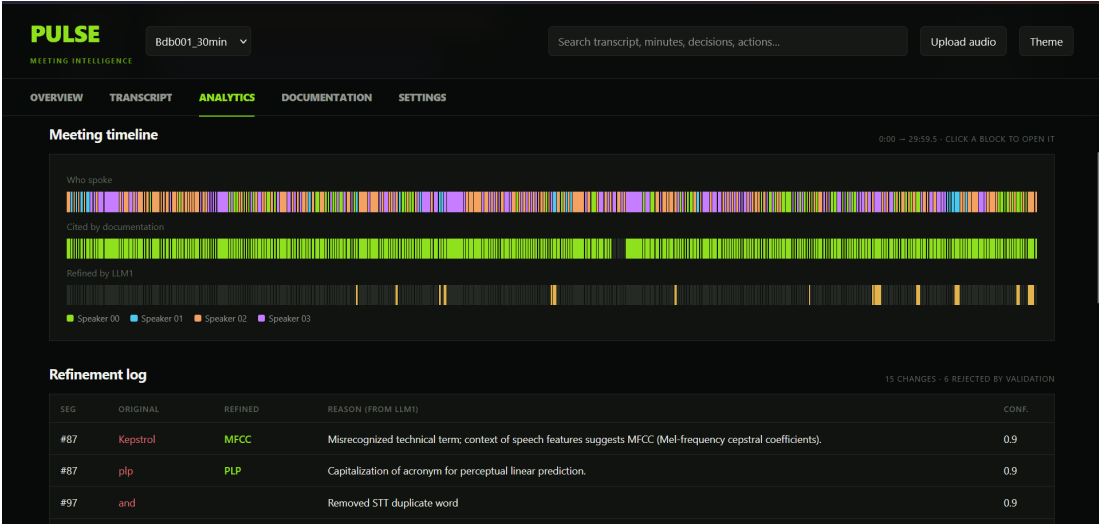


Figure 4: Analytics tab with the speaker timeline and refinement log.

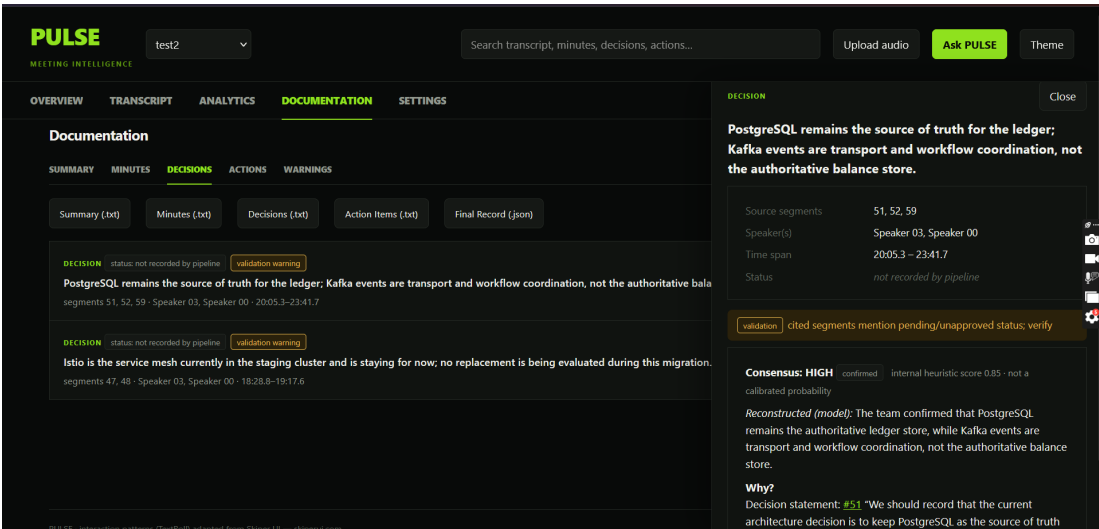


Figure 5: Documentation tab with the evidence panel for a decision.

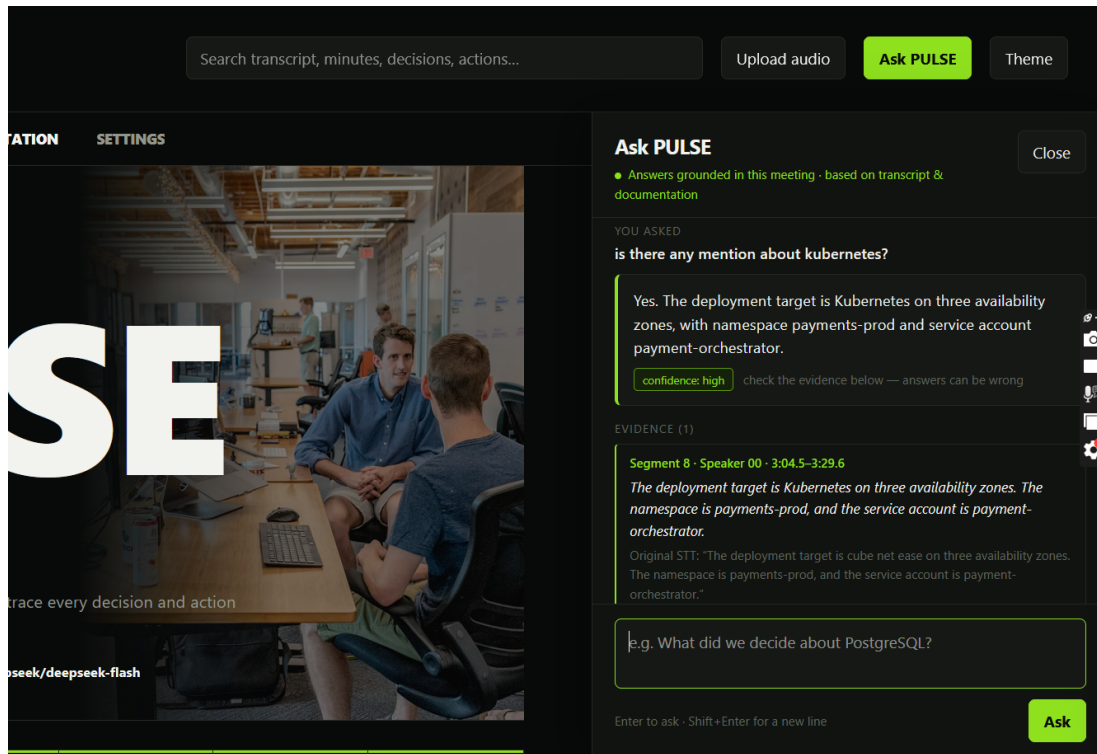


Figure 6: Ask PULSE answering a question with cited evidence.

References

- [1] V. Rennard, G. Shang, J. Hunter, and M. Vazirgiannis, "Abstractive Meeting Summarization: A Survey," *Transactions of the Association for Computational Linguistics*, vol. 11, 2023.
- [2] Z. Du, S. Zhang, S. Zheng, and Z.-J. Yan, "Speaker Overlap-aware Neural Diarization for Multi-party Meeting Analysis," *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 7458–7469, 2022.
- [3] Y. Jin, Q. Shi, and Q. Liu, "CFAS: Consensus-Focused Abstractive Meeting Summarization through Multi-Party Discourse Modeling," *Journal of King Saud University – Computer and Information Sciences*, vol. 37, art. 187, 2025.
- [4] S. Inoue, H. Nguyen, H. Pham, T. Liu, and M.-T. Nguyen, "Meeting Decision Tracker: Making Meeting Minutes with De-Contextualized Utterances," *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the ACL and the 12th International Joint Conference on Natural Language Processing: System Demonstrations*, pp. 88–94, 2022.